

IK 515 : Music and Cognition

Course Code : IK 515

Course Name : Music and Cognition

L-T-P-C : 3-0-0-3

Intended for : BTech/MTech/MS/MSc/MA/Ph.D

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Introduction to Music and Cognition:** Basics of cognitive science and musical structure, exploring the ways music and cognitive processes interact, and the role of music in cognitive enhancement and therapy. (5 Hours)
- **Musical Perception:** Delve into the cognitive processing of musical elements such as pitch, rhythm, timbre, and melody. Explore the auditory system, musical feature extraction, and the cognitive organization of musical sounds. (10 Hours)
- **Musical Memory:** Understand the intricacies of short-term and long-term musical memory and musical expectation. Examine the encoding, storage, and retrieval of musical information. (8 Hours)
- **Emotion and Music:** Examination of how music evokes emotional responses, the role of musical expression, and the neuroscientific basis of musical emotions. Understand the role of cultural and individual differences in musical emotion. (9 Hours)
- **Music, Intelligence, and Learning:** Analysis of the Mozart Effect, exploration of the impact of musical training on cognitive development, and the relationship between music and spatial-temporal reasoning. (10 Hours)

Laboratory/practical/tutorial Modules:

1. NA

Textbooks:

1. Levitin, D. J., **This Is Your Brain on Music: The Science of a Human Obsession**, Penguin, 2007.
2. Sloboda, J. A., **The Musical Mind: The Cognitive Psychology of Music**, Oxford University Press, 1985.

References:

1. Patel, A. D., **Music, Language, and the Brain**, Oxford University Press, 2008.
2. Hodges, D. A., **Handbook of Music Psychology**, IMR Press, 1996.